Lecture 5 Stock Paths as Geometric Brownian Motion

Review of the Binomial Random Walk

We previously considered an n-step binomial tree model of a random walk:

$$S(0), S(1), \ldots, S(n),$$

where:

$$S(n) = r_1 + r_2 + \cdots + r_n,$$

and each r_i is a discrete return:

$$r_i = \begin{cases} u & \text{with probability } p, \quad u > 0, \\ d & \text{with probability } 1-p, \quad d < 0. \end{cases}$$

The ratio of prices at two consecutive time steps is:

$$\frac{S(i)}{S(i-1)} = \begin{cases} 1+u & \text{with probability } p, \\ 1+d & \text{with probability } 1-p. \end{cases}$$

Counting Up and Down Moves

We introduce the following notations:

- $H(n)$: the number of times the walk moves up.
- $T(n)$: the number of times the walk moves down.

Thus, the stock price after n steps can be written as:

$$\frac{S(n)}{S(0)} = \prod_{i=1}^{n} \frac{S(i)}{S(i-1)} = (1+u)^{H(n)}(1+d)^{T(n)}.$$

对称随机游走 Symmetric Random Walk

In the special case of a symmetric random walk, we have:

$$r_i = \begin{cases} 1 & \text{with probability } \frac{1}{2}, \\ -1 & \text{with probability } \frac{1}{2}. \end{cases}$$

Define:

$$M_n = r_1 + \cdots + r_n.$$

For M_n, the mean and variance are:

$$\mathbb{E}[M_n] = 0, \quad \mathrm{Var}(M_n) = \sum_{i=1}^{n} \mathrm{Var}(r_i) = n.$$

尺度化的对称随机游走 Scaled Symmetric Random Walk

To move toward continuous time, we define a scaled version of the symmetric random walk.

Let $t > 0$, $t \in \mathbb{Q}$, and select $n \in \mathbb{N}$ such that $nt \in \mathbb{N}$. Define:

$$W_n(t) := \frac{1}{\sqrt{n}} M_{nt}.$$

This represents the symmetric random walk with nt steps, scaled by $\frac{1}{\sqrt{n}}$.

M_{nt} 是对称随机游走的总和:

$$M_{nt} = r_1 + r_2 + \cdots + r_{nt}$$

其中每个 $r_i \in \{+1, -1\}$，概率各为 $\frac{1}{2}$。 M_{nt} 的尺度随着 n 增大也增大，不是一个“稳定”的过程。

我们希望构造一个过程 $W_n(t)$，它在 $n \to \infty$ 时收敛到布朗运动 $W(t)$。

- 由于:

$$\mathrm{Var}(M_{nt}) = nt$$

所以:

$$\mathrm{Var}\left(\frac{1}{\sqrt{n}} M_{nt}\right) = \frac{nt}{n} = t$$

So we need to 除以 $\sqrt{n}$。我们定义:

$$W_n(t) := \frac{1}{\sqrt{n}} M_{nt}$$

是因为它:

- 让离散随机游走的方差稳定在 t,
- 使得 $W_n(t) \xrightarrow{d} \mathcal{N}(0, t)$,
- 是布朗运动构造的标准步骤之一（也是 Donsker 定理的核心形式）。

The expectation and variance of $W_n(t)$ are:

$$\mathbb{E}[W_n(t)] = 0,$$
$$\mathrm{Var}(W_n(t)) = \frac{1}{n} \cdot \mathrm{Var}(M_{nt}) = \frac{1}{n} \cdot nt = t.$$

Application of the Central Limit Theorem

By the Central Limit Theorem, as $n \to \infty$, we have:

$$\lim_{n \to \infty} W_n(t) \sim \mathcal{N}(0, t),$$

which means that $W_n(t)$ converges in distribution to a normal random variable with mean 0 and variance t. Equivalently, we may write:

$$W_n(t) \xrightarrow{d} \sqrt{t} \cdot \mathcal{N}(0, 1).$$

Thus, the limit distribution is the normal distribution with mean 0 and standard deviation $\sqrt{t}$.

Tabelle 1: 离散模型与连续模型的关系

离散模型	通过缩放和极限定理	得到连续模型
二项树模型	$n \to \infty$	几何布朗运动（股票价格模型核心）
对称随机游走	缩放后应用 CLT	标准布朗运动（连续时间极限）

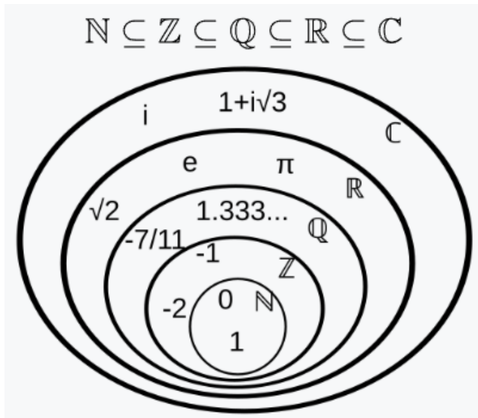

Abbildung 1: Hierarchy of Number Sets

Problem Setup for Continuous Limit

Let $t \in \mathbb{Q}_{>0}$（一个正有理数）.

We are interested in the distribution of stock price $S(t)$ at time t（股票价格在时间 t 的分布）, starting from $S(0)$. In particular, we assume:

1. For each $n \in \mathbb{N}$ such that $nt \in \mathbb{N}$, there is an nt-step binomial walk $S_n(t)$ approximating $S(t)$. 让二项树模型 $S_n(t)$ 去逼近连续模型 $S(t)$。
2. 假设均值为 0（暂时没有漂移 drift). The process has zero drift:
$$\mathbb{E}[S(t) - S(0)] = \mathbb{E}[S_n(t) - S(0)] = 0.$$
(Remark: For now, we assume zero drift. The inclusion of drift will be addressed later.)
3. There exists $\sigma > 0$ such that for each n:
$$\mathrm{Var}(S_n(t)) = \sigma^2 t.$$
4. At each step of $S_n(t)$, we assign probability $\frac{1}{2}$ for moving up or down.
$$S(t) = S(0) \exp\left(-\frac{1}{2}\sigma^2 t + \sigma\sqrt{t} \cdot \mathcal{N}(0,1)\right),$$
where $\mathcal{N}(0,1)$ is a standard normal distribution.

Proof Sketch

We begin with:

$$S_n(t) - S(0) = \hat{r}_{n,1} + \hat{r}_{n,2} + \cdots + \hat{r}_{n,nt},$$

where

$$\hat{r}_{n,i} = \begin{cases} u_n & \text{with probability } \frac{1}{2}, \\ d_n & \text{with probability } \frac{1}{2}. \end{cases}$$

By assumption:

$$\mathbb{E}[S_n(t) - S(0)] = 0 \implies \mathbb{E}[\hat{r}_{n,i}] = 0.$$

For variance:

$$\mathrm{Var}(S_n(t) - S(0)) = \mathrm{Var}(S_n(t)) = \sigma^2 t.$$

Since the $\hat{r}_{n,i}$ are i.i.d., we have:

$$\mathrm{Var}(S_n(t)) = nt \cdot \mathrm{Var}(\hat{r}_{n,i}).$$

Thus,

$$\mathrm{Var}(\hat{r}_{n,i}) = \frac{\sigma^2}{n}.$$

Binary Representation of Steps

By some algebra, we deduce:

$$\hat{r}_{n,i} = \begin{cases} \frac{\sigma}{\sqrt{n}} & \text{with probability } \frac{1}{2}, \\ -\frac{\sigma}{\sqrt{n}} & \text{with probability } \frac{1}{2}. \end{cases}$$

也就是说，单步涨跌幅度随着 n 增大而收敛到 0（因为时间步长越来越短）。

Counting Up and Down Moves

Let:

- $H_n(t)$ be the number of up moves.
- $T_n(t)$ be the number of down moves.

The total number of steps satisfies:

$$H_n(t) + T_n(t) = nt$$

$$H_n(t) - T_n(t) = M_{nt}$$

其中 M_{nt} 就是前面定义的对称随机游走累积和。

- 于是:

$$H_n(t) = \frac{nt + M_{nt}}{2}, \quad T_n(t) = \frac{nt - M_{nt}}{2}$$

Thus,

$$\frac{S_n(t)}{S(0)} = (1 + u_n)^{H_n(t)} (1 + d_n)^{T_n(t)}.$$

Since:

$$u_n = \frac{\sigma}{\sqrt{n}}, \quad d_n = -\frac{\sigma}{\sqrt{n}},$$

we have:

$$\frac{S_n(t)}{S(0)} = \left(1 + \frac{\sigma}{\sqrt{n}}\right)^{H_n(t)} \left(1 - \frac{\sigma}{\sqrt{n}}\right)^{T_n(t)}.$$

Expressing $H_n(t)$ and $T_n(t)$

We have:

$$H_n(t) = \frac{nt + M_{nt}}{2}, \quad T_n(t) = \frac{nt - M_{nt}}{2},$$

where M_{nt} is the symmetric random walk sum after nt steps.
Thus,

$$\frac{S_n(t)}{S(0)} = \left(1 + \frac{\sigma}{\sqrt{n}}\right)^{\frac{nt+M_{nt}}{2}} \left(1 - \frac{\sigma}{\sqrt{n}}\right)^{\frac{nt-M_{nt}}{2}}.$$

Taking logarithms:

$$\ln\left(\frac{S_n(t)}{S(0)}\right) = \frac{nt + M_{nt}}{2} \ln\left(1 + \frac{\sigma}{\sqrt{n}}\right) + \frac{nt - M_{nt}}{2} \ln\left(1 - \frac{\sigma}{\sqrt{n}}\right).$$

Taylor Expansion

We apply the Taylor expansion:

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \mathcal{O}(x^4),$$

where $\mathcal{O}(x^A)$ denotes error terms such that for any $\delta > 0$:

$$\lim_{x \to 0} \frac{\mathcal{O}(x^A)}{x^{A-\delta}} = 0.$$

This allows us to simplify the expression and transition to the continuous-time limit.

Taylor Expansion to Derive the Limit

We apply the Taylor expansion:

$$\ln(1+x) = x - \frac{1}{2}x^2 + \mathcal{O}(x^3).$$

由于 $x = \frac{\sigma}{\sqrt{n}} \to 0$，高阶项可以忽略，只保留前两项。
Thus, we rewrite:

$$\ln\left(\frac{S_n(t)}{S(0)}\right) = \frac{nt+M_{nt}}{2}\left(\frac{\sigma}{\sqrt{n}} - \frac{1}{2}\frac{\sigma^2}{n} + \mathcal{O}(n^{-3/2})\right) + \frac{nt-M_{nt}}{2}\left(-\frac{\sigma}{\sqrt{n}} - \frac{1}{2}\frac{\sigma^2}{n} + \mathcal{O}(n^{-3/2})\right).$$

After simplifying terms:

$$\ln\left(\frac{S_n(t)}{S(0)}\right) = -\frac{t\sigma^2}{2} + \frac{\sigma}{\sqrt{n}}M_{nt} + \mathcal{O}(n^{-1/2}) + M_{nt}\cdot\mathcal{O}(n^{-3/2}).$$

Recalling that:

$$\frac{M_{nt}}{\sqrt{n}} = W_n(t),$$

we obtain:

$$M_{nt} + \mathcal{O}(n^{-1/2}) = W_n(t)\cdot\mathcal{O}(n^{-1}) \to 0 \quad \text{as } n\to\infty.$$

Taking the Limit

Thus, taking the limit as $n \to \infty$:

$$\lim_{n\to\infty}\ln\left(\frac{S_n(t)}{S(0)}\right) = -\frac{t\sigma^2}{2} + \sigma W(t),$$

where $W(t)$ is standard Brownian motion with:

$$W(t) \sim \mathcal{N}(0,t).$$

Exponentiating both sides gives:

$$S(t) = S(0)\exp\left(-\frac{\sigma^2}{2}t + \sigma\sqrt{t}\cdot\mathcal{N}(0,1)\right).$$

Drift and Interest Considerations

Interest: The time value of money is represented by interest rate r, where:

$$\$X \quad \text{grows to} \quad e^{rt}X \quad \text{over time } t.$$

Distribution of $S(t)$: The discounted stock price is:

$$e^{-rt}S(t).$$

Thus,

$$S(t) = S(0)\exp\left(-rt - \frac{\sigma^2}{2}t + \sigma\sqrt{t}\cdot\mathcal{N}(0,1)\right).$$

Including drift μ: If μ is the drift of log-returns, then:

$$S(t) = S(0)\exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma\sqrt{t}\cdot\mathcal{N}(0,1)\right).$$

This is the **Geometric Brownian Motion (GBM)** formula.

Generalized Brownian Motion Definition

When $nt \notin \mathbb{N}$, we define $W_n(t)$ via interpolation:

$$W_n(t) = (1-q)\frac{1}{\sqrt{n}}M_{\lfloor nt\rfloor} + q\frac{1}{\sqrt{n}}M_{\lfloor nt\rfloor+1},$$

where:

$$q = nt - \lfloor nt\rfloor.$$

We have:

$$\mathbb{E}[W_n(t)] = 0, \quad \lim_{n\to\infty}\mathrm{Var}[W_n(t)] = t,$$

and thus:

$$\lim_{n\to\infty} W_n(t) \sim \mathcal{N}(0,t).$$